

PTSG-Core — Command × Execution-Phase Behavior Table (normative, v1.1, 2026-07)									
Command	Op / Sub-op	Phase	Stay Counter	Timing Signals	Prescaler Tick	State Number	Edge	Legality	Notes
Stay	opcode 1	FG	Continues counting (no reset)	Driven (changes per instruction)	Consumes ticks (waits)	Increments on the tick at stay-count expiry (Stay-timeup)	Trailing	Legal	The wait itself: SN rests here until Stay-timeup = the Nth prescaler tick after Stay Set (C4-F10, corrected). Queued reservations fire at this trailing edge (C4-F11).
		BG	—	—	—	—	—	N/A	There is no background Stay: a Stay reached by the window scan IS the FG Stay — the forced end of the window content and the start of the next stay period.
		Q	—	—	—	—	—	N/A	Likewise: a Stay reached in the Q band is the FG Stay — the end of the queue-reservation region and the start of the next stay period.
Branch	opcode 2	FG	No effect (already 0)	Driven (changes per instruction)	Consumes one tick (waits for next)	Waits for the tick → Condition true: increments; Condition fail: updated to operand + current State Number	Trailing	Legal	
		BG	Continues counting (no reset). On-Tick	Held (unchanged / frozen)	Ignored (full system clock)	On the next clock → Condition true: increments; Condition fail: updated to operand + current State Number	Trailing	Legal	
		Q	Continues counting (no reset). On-Tick	Held (unchanged / frozen)	Ignored (full system clock)	Increments on the next clock (simultaneously registers the operand as the queued target)	Trailing	Legal	Condition is evaluated at firing (Stay-timeup), when the queued target executes; the relative State Number is computed then.
Jump	opcode 3	FG	No effect (already 0)	Driven (changes per instruction)	Consumes one tick (waits for next)	Waits for the tick, then updated to the address given by the operand	Trailing	Legal	
		BG	Continues counting (no reset). On-Tick	Held (unchanged / frozen)	Ignored (full system clock)	Updated to the operand address on the next clock	Trailing	Legal	The window scan itself can be redirected.
		Q	Continues counting (no reset). On-Tick	Held (unchanged / frozen)	Ignored (full system clock)	Increments on the next clock (simultaneously registers the operand as the queued target)	Trailing	Legal	
Global · Reset	sub-op 0	FG	No effect (already reset)	Driven (changes per instruction)	Ignored (full system clock)	Set to 0 on the next clock	LEADING ⚠	Legal	Leading-edge exception (Trailing-Edge Doctrine, Ch1 §1.4a); depends on the preceding command ending on a prescaler tick. Execution bands: §3.4a (PROVISIONAL). Never resets presc_cnt (C3-F21).
		BG	Arms (reset to 0, don't start)	Driven (changes per instruction)	Ignored (full system clock)	Set to 0 on the next clock	Trailing	Legal	"Staff meal" band — emergencies only (§3.4a, PROVISIONAL). Never resets presc_cnt (C3-F21).
		Q	Arms (reset to 0, don't start)	Driven (changes per instruction)	Ignored (full system clock)	Increments on the next clock (at queued execution: set to 0)	Trailing	Legal	Effectively prescaled — fires at Stay-timeup (§3.4a). Timing signals stay Driven so a "Reset reserved" notification can be emitted, set them equal to the Stay's value if no distinct output is wanted. Never resets presc_cnt (C3-F21).
Global · Base Set	sub-op 1	FG	No effect (untouched)	Held	—	Held (HALT)	Trailing	HALT	Error HALT (runaway detection, C3-F24); window-only command in the foreground (FG-Global exclusion, C3-F23).
		BG	Continues counting (no reset). On-Tick	Held (unchanged / frozen)	Ignored (full system clock)	Increments on the next clock (simultaneously loads Base := current State Number)	Trailing	Legal	Space-axis loop base. If this Base Set's matching Loop has not been exited by the time Prog End is reached → Error HALT (band-pairing check, C3-F24).
		Q	Continues counting (no reset). On-Tick	Held (unchanged / frozen)	Ignored (full system clock)	Increments on the next clock (simultaneously loads Base := Stay Start State register)	Trailing	Legal	Time-axis loop base: a queued loop must return to the start of the stay period, named by the Stay Start State register (C3-F25). If the matching Loop lies inside the BG window → Error HALT (band-pairing check).
Global · Stay Set	sub-op 2	FG	Arms (reset to 0) and starts counting	Driven (changes per instruction)	Ignored (full system clock)	Increments on the next clock	LEADING ⚠	Legal	Leading-edge exception (Ch1 §1.4a). Counting is On-Tick from here through the window; the Stay instruction never clears it (C4-F10, corrected). Also writes its own State Number to the Stay Start State register — valid within this stay cycle, until a queued Base Set hands it to
		BG	Continues counting (WITH reset) — re-kick	Held (unchanged / frozen)	Consumes one tick (waits for next)	Waits for the tick, then increments	Trailing	Legal	The re-kick: re-arms the running count mid-window (variable-length Stay), tick-synchronized. Not yet in RTL (CHANGES C4).
		Q	Continues counting (no reset). On-Tick	Held (unchanged / frozen)	Ignored (full system clock)	Increments on the next clock	Trailing	Legal	Registered during the scan; firing semantics to be exercised in verification (CHANGES C4).
Global · Return	sub-op 3	FG	No effect (untouched)	Held	—	Held (HALT)	Trailing	HALT	Error HALT (runaway detection, C3-F24); window-only command in the foreground (C3-F23).
		BG	Continues counting (no reset). On-Tick	Held (unchanged / frozen)	Ignored (full system clock)	Set to the Return State Number on the next clock	Trailing	Legal	
		Q	Continues counting (no reset). On-Tick	Held (unchanged / frozen)	Ignored (full system clock)	Increments on the next clock (at queued execution → Return State Number)	Trailing	Legal	
Global · Sub-sequence Call	sub-op 4	FG	No effect (untouched)	Held	—	Held (HALT)	Trailing	HALT	Error HALT (runaway detection, C3-F24); window-only command in the foreground (C3-F23).
		BG	Continues counting (no reset). On-Tick	Held (unchanged / frozen)	Ignored (full system clock)	Set to Sub-sequence offset + current State Number on the next clock	Trailing	Legal	D16–D31 is unambiguously the operand here: timing signals are Held in this band — the operand/timing-signal dual-use question is dissolved by banding itself (C3-F23).
		Q	Continues counting (no reset). On-Tick	Held (unchanged / frozen)	Ignored (full system clock)	Increments on the next clock (at queued execution → Sub-sequence offset + current State Number)	Trailing	Legal	
Global · Loop	sub-op 5	FG	No effect (untouched)	Held	—	Held (HALT)	Trailing	HALT	Error HALT (C3-F23/F24). FG timing-sequence loops use the queue instead. Because the loop-count operand occupies D16–D31, loop_cnt_match cannot be a timing signal; at queued execution it can be emitted at clock width.
		BG	Continues counting (no reset). On-Tick	Held (unchanged / frozen)	Ignored (full system clock)	On the next clock → within the loop count: set to Base register; count reached: increments	Trailing	Legal	Looping within the BG program window (space axis).
		Q	Continues counting (no reset). On-Tick	Held (unchanged / frozen)	Ignored (full system clock)	Increments on the next clock (at queued execution → within the loop count: set to Base register; count reached: increments)	Trailing	Legal	FG timing-sequence loops use the queue (time axis). With a queued Base Set this returns to the stay-period origin — the single-period self-loop yields the 2'28 pattern: StaySet → ProgEnd → BaseSet(Q) → Loop-65536(Q) → Stay-4096 (C3-F25).
Global · Prog End	sub-op 6 (tentative)	FG	No effect (untouched)	Held	—	Held (HALT)	Trailing	HALT	Error HALT if encountered outside an open background-program window (runaway detection; supersedes the earlier "blank shot / no effect" reading).
		BG	Continues counting (no reset). On-Tick	Held (unchanged / frozen)	Ignored (full system clock)	Increments on the next clock	Trailing	Legal	This command IS the band divider: before it = BG (immediate band), after it = Q (reservation band) (C3-F2).
		Q	No effect (untouched)	Held	—	Held (HALT)	Trailing	HALT	A second Prog End in the Q band → Error HALT (runaway detection).
Global · NOP	sub-op 7	FG	No effect (already 0)	Driven (changes per instruction)	Consumes one tick (waits for next)	Waits for the tick, then increments	Trailing	Legal	State-0 cold-start absorber (C4-V3) when placed at state 0.
		BG	Continues counting (no reset). On-Tick	Held (unchanged / frozen)	Ignored (full system clock)	Increments on the next clock	Trailing	Legal	
		Q	Continues counting (no reset). On-Tick	Held (unchanged / frozen)	Ignored (full system clock)	Increments on the next clock	Trailing	Legal	

Legend: red rows = Error HALT (runaway detection, C3-F24); amber Edge cells (⚠) = sanctioned leading-edge exceptions (FG Stay Set / Reset, Ch1 §1.4a); "On-Tick" = the stay counter increments on prescaler ticks while the window is open (C4-F10, corrected). Full definitions: Read Me sheet and repository Chapter 3 §3.4b.